

Unit 1: Introduction

Computer:

The term computer is taken from Greek work “Computare” which means “to calculate”. Computer is an advanced programming and data processing electronic device that takes raw data as input from the user and processes these data under the control of set of instructions and gives the result and saves output for the future use. It can process both numerical and non-numerical calculations. A computer can process data, pictures, sound and graphics. They can solve highly complicated problems quickly and accurately.

Computer is a programmable electronic device which is capable of storing and receiving data from input device and performs a sequence of operations in according to the predetermined set of instructions to produce a result in the form of information or signals.

A computer performs basically five major computer operations or functions irrespective of their size and make. These are

- it accepts data or instructions by way of input,
- it stores data,
- it can process data as required by the user,
- it gives results in the form of output, and
- It controls all operations inside a computer.

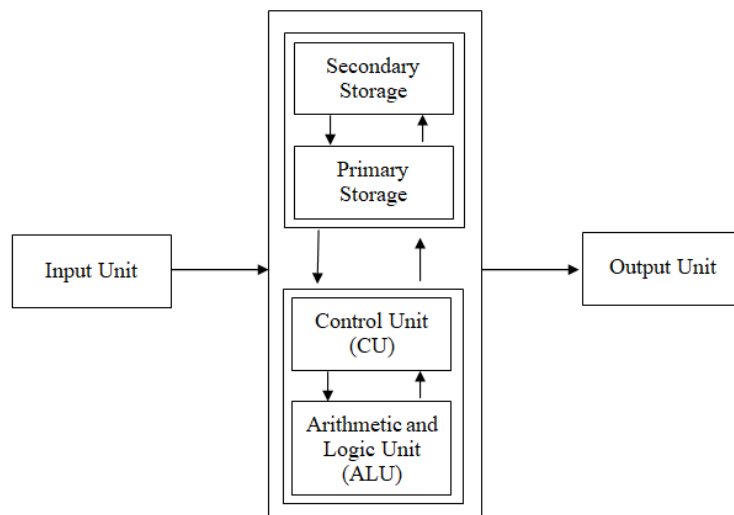


Fig: Block diagram of computer

Computers are used in varieties of different areas. We can't imagine this 21st century without computer. Some major application areas of computer are as follows:

a) School / College / Universities

Computers are widely used in school, college or universities to perform following task:

- To keep records of individual students such as affiliated year, course registered, mark ledgers, etc
- To keep individual records of teachers, lecturers, or other faculty members.
- Used as a learning tools. A student / teacher can collect (browse) various information about their study
- New method of online / offline learning such as e-learning, m-learning, distance learning, CAVE (Cave Automatic Virtual Environment) etc

b) Bank

All financial institutions (bank) uses computer widely to perform following task:

- To keep individual records of their employees and clients

- To perform general purpose tasks such as: to maintain ledger, to make entry in passbooks, daily transactions.
- To provide online services to their customers
- To provide ATM (Automated Teller Machine) facility
- To provide several services such as EFT (Electronic Fund Transfer) ABBS (Any Branch Banking Service) service, mobile banking facilities, etc

c) **Offices**

Computers are an essential part of office management. Computers are widely used in office for:

- Computers can be used to generate data. Employees can use the Internet to find current information that the company needs to operate efficiently.
- One of the most common uses of computers in an office setting is to communicate with others inside and outside the workplace.
- It helps in keeping of records and therefore it serves as a substitute of filing.
- It is helpful for analyzing, comparing and evaluating various projects

d) **Health and Medicine**

Computers have become an important part in hospitals, labs, and dispensaries. They are used:

- to keep the history of patients and medicines
- It is used as a research tool by the doctors
- A junior doctor uses it to consult with his senior specialist doctor
- It is also used in scanning and diagnosing different diseases.
- As a virtual surgery tool.

e) **Communication**

Now a day, computers are frequently used by an individuals or organization:

- As a means of faster and cheaper communication media.
- To communicate with each other's using any forms of communication: text, audio, video, and graphics at high speed and minimum communication cost.
- We can use social sites such as face book, twitter, etc to express our views, share photos and many more emotions to the society
- Video conferencing is an advanced method of communication through which we can communicate verbally throughout the world.

f) **Simulation:**

Simulation is an experiment to model a real life or hypothetical situation through computer so that it can be studied to view how the system works i.e. it is a tool to virtually investigate the behavior of the propose system. Now days the computers are widely used as simulation tool for:

- The medical field uses simulations to train practitioners in a multitude of skills and environments.
- It also provides students the opportunity to develop communication and teamwork skills.
- Business simulations offer an immersive environment in which marketing strategies and investment opportunities can be practiced without the expense and risk of capital outlay.
- Demanding customers and changing markets conditions can be manipulated to afford practice in both customer relations as well as economic processes.
- It also play an important role in the training of weapons instruction to interpersonal skills,

g) **Business**

- Payroll Calculations
- Budgeting
- Sales Analysis

- Financial forecasting
- Managing employees database
- Maintenance of stocks, etc.

h) Transportation

The computers are used in transportation filed to perform following activities:

- For online reservation of different vehicles.
- Monitoring system for temperature, pressure, and flow rates and so on.
- Training and simulation systems that are used by crew to ensure safe operation of the vehicles.

i) Entertainment

Now a day's computers are widely used as an entertainment tools. Using computer:

- We can watch movies, listen to songs and watch videos, download different stuff.
- We can also watch live matches on the internet.

j) Home

Computers are also used by an individual to perform various activities at home. Some of them are:

- Used to manage home budget.
- Used as recreating tools such as playing games, watching videos, listening music, etc.
- Used to manage the office work at home i.e. owner of a company can check the work of the employees from home.
- To collect different types of information's related to our study
- To chat and share videos, photos, etc with friends and family using internet facility

Generation of Computer:

Each generation of computer is characterized by a major technological development that fundamentally changed the way computers operate, resulting in increasingly smaller, cheaper, and more powerful and more efficient and reliable devices. The various generations of computers are listed below:

Generation of Computer:

1. First Generation Computer:

- Time Duration : 1945 to 1954
- Main component : Vacuum tubes (Lee De Forest)
- Power consumption : Maximum power consumption
- Processing Speed : Very Slow (Milli seconds)
- Size : Huge and Bulky (30 – 32 tons)
- Memory capacity : Limited (Bytes, Kilo Bytes)
- Programming Language : Machine Level language
- Cost : Very Expensive
- Example : ENIAC (Electronic Numerical And Integrated Automatic Computer), EDVAC (Electronic Discrete Variable Automatic Computer), EDSAC (Electronic Delay Storage Automatic Computer) UNIVAC (Universal Automatic Computer), IMB 701 etc

2. Second Generation Computer:

- Time Duration : 1955 to 1964
- Main component : Transistor (William Shockley, John Bardeen & Walter Brattain)
- Power consumption : Less power consumption
- Processing Speed : Smaller as compared to 1st generation (Micro seconds)
- Size : Huge

- Memory capacity : Slight high (Mega bytes)
- Primary storage : Magnetic core memory
- Programming Language : Low Level Languages
- Cost : Expensive
- Used for : commercial and scientific applications
- Example : IBM 1401, Honeywell 400, CDC 1604, ICL 1901 etc

3. Third Generation Computer:

- Time Duration : 1965 to 1974
- Main component : IC (Robert Noyce, J.C. Kilby)
- Heat consumption : Less
- Processing Speed : Fast (Nano Seconds)
- Size : Medium
- Memory capacity : High (Giga Bytes)
- Programming Language : Low level languages, High Level language
- Cost : Less expensive
- Additional Feature:
 - ✓ Introduction of TUI based Operating System
 - ✓ Large magnetic core memory
 - ✓ More flexible input/output devices
 - ✓ Mini computers were emerged
- Example : IBM 360 / 370, CDC 6600, PDP 11, Honeywell 2200, ICT 1900, etc

4. Fourth Generation Computer:

- Time Duration : 1975 to till now
- Main component : Microprocessor (VLSI)
- Heat consumption : Very less
- Processing Speed : Fast (Pico seconds)
- Size : Very small (Desktop, Laptop and Palmtop)
- Memory capacity : Very huge (100 GB, 500 GB, 1TB)
- Programming Language : High Level languages & 4 GL
- Cost : Cheaper (Affordable by common user)
- Additional features:
 - ✓ GUI based operating system were introduced
 - ✓ Semiconductor memories were replaced for magnetic core memories
 - ✓ Computer network were developed
 - ✓ e-mail, internet services was introduced
- Example : IBM PC, Apple, Motorola M6800, Burroughs B7700, etc

5. Fifth Generation Computer:

- Time Duration : under construction
- Main component : Bio-chips / ULSI
- Heat consumption : Very low
- Processing Speed : very high
- Memory capacity : very high
- Programming Language : Natural language
- Additional features:
 - ✓ Concept of AI (Artificial Intelligence) i.e. thinking as a human being.
 - ✓ Voice recognition, pattern recognition, hand writing recognition

Types of Computer:

Classification of computer is shown as following diagram.

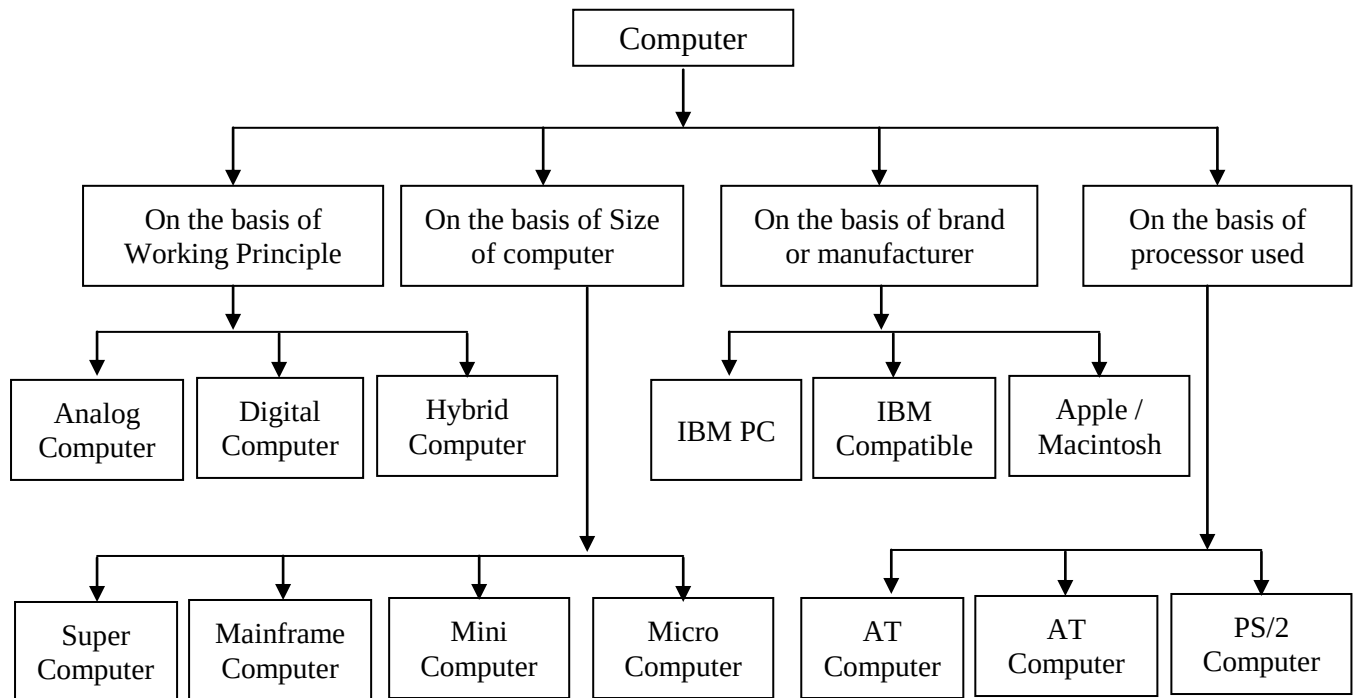


Fig: - Classification of computer

1. On the basis of working principle

There are three types of computer and they are:

a) Analog computer

- The computers which performs the functions on continuously variation in inputs which can be in form of fluctuation in voltage or temperature, electrical or mechanical parts are called analog computer
- The computers are used to measure the electrical signal, current, frequency of signal and voltage, resistance of capacitor, etc
- It has low memory and slow speed
- It is less reliable and used for special purpose.
- Example: Plesley

b) Digital computer

- The computers which present physical quantities with the help of symbols or numbers and provide us discrete information are called digital computer
- Accuracy rate of such computes is very high as compared to analog computer.
- It has very large storage capacity with high speed
- It is more reliable and used for general purpose
- Example: personal computers, laptops, smart phone etc.

c) Hybrid computer

- The computer that exhibit features of both analog and digital computers are called hybrid computer.
- It is a digital computer that accepts analog signal, converts them to digital and processes them in digital form.
- It has the speed of analog and accuracy of digital computer.

- Generally used in scientific application or in controlling industrial processes., hospitals to measure the heartbeat of the patients, etc.

2. On the basis of size of a computer

Computers are also categorized on the size of the computers. There are four types of computers and they are:

a) Super computer

- The most powerful computers in terms of performance and data processing and data storage capacity.
- Very expensive and large in size that is designed to perform specific tasks.
- Generally used by large organizations and government agencies for various purposes such as: weather forecasting, space exploration, missile and weapon controls, etc.
- Example: Tianhe – 2 (Milky Way – 2) (China), Titan, Cray Inc. (United States), Sequoia, IBM (United States), K Computer, Fujitsu (Japan), Mira, IBM, (United States)

b) Mainframe computer

- It is also large sized computer which occupies large room area.
- The computers are powerful enough to support several hundred usea simultaneousl at remot terminals.
- It can process and store large amount of data.
- Generally bank, educational institutions, insurance companies etc uses such computer to store data about their customers and stakeholders.
- Example: IBM s/390, Amdahl 580, Control Data Cyber 176, Fujitsu's ICL VME, Hitachi's Z800, etc

c) Mini computer

- The computers are also multi user systems like mainframe computers.
- Such computers are used by small business firms but not for single user.
- Also known as Midrange computers or medium sized computers.
- Example: IBM AS / 400, IBM SYSTEM 360, HP 3000, PRIME 9755,K-202, Texas Instrument TI – 990, SDS – 92, IBM Midrange computers, etc

d) Micro computer

- The computers are cheapest and smallest in size among other computers.
- Such computers are specially designed for general usage like entertainment, education and various personal uses.
-

3. On the basis of brand or manufacturer

The computers are also categorized according to the bran or manufacturer of computer.

a) IBM PC

- The computes which are manufactured by IBM Company itself are called IBM PC
- Such computers are also known as branded computers.
- The computers are more reliable, durable and have better quality.
- The computers are too costly.
- The computers are specifically designed for professionals and scientific problem solvers.

b) IBM Compatible

- The computers those are similar to the original IBB PC are known as IBM Compatible
- The computers are also known as PC clones or IBM Clones.
- The computers are easily available and less expensive as compared to IBM PC.

c) Apple / Macintosh

- The Computers made by Apple Corporation its self are known as Apple Computers.
- The computers are also known as Mac Computer.
- Such computers use their own software and hardware.
- Software developed for IBM computer can't run in Apple and vice-versa.
- The computers are totally different than that of IBM's and are very costlier.

4. On the basis of processor used

There are three types of computer and they are:

a) XT Computer

- It stands for Extended Technology Computer.
- The computer which have CPU of 8080, 8086, or older version is known as XT Computer
- It cannot support GUI based operating system
- The input / output device are non flexible

b) AT Computer

- It stands for Advanced Technology computer.
- The computer which have CPU of 80286 or higher version of CPU is known as AT Computer.
- It can support both GUI and CUI based operating System.
- The input / output device are very flexible.

c) PS/2 Computer

- It is an advancement of AT computer.
- Generally such computers are rechargeable and battery based system.
- It is operated with OS/2 operating system.

Software:

A computer cannot do anything on its own. It must be instructed to do a desired job. Hence, it is necessary to specify a sequence of instructions, which a computer must perform to solve a problem. Such a sequence of instructions, written in a language, which can be understood by a computer, is called a computer program. It is the program, which controls the activity of processing by the computer, and the computer performs precisely what the program wants it to do. When a computer is using a program, we say, it is running or executing that program. The term software is refers to the set of computer programs, procedures, and associated documents (flowcharts, manuals, etc.), which describes the programs, and how they are to be used. To be precise, software means a collection of programs, whose objective is to enhance the capabilities of the hardware. A software package is a group of programs, which solve a specific problem or perform a specific type of job. For example, a word-processing package may contain programs for text editing, text formatting, drawing graphics, spelling checking, etc. Hence, a multipurpose computer system has several software packages, one each for each for every type of job it can perform.

Types of software

Although, the range of software available today is vast and varied, most software can be divided into two major categories:

- System Software,
- Application Software

1. System Software:

System software is a set of one or more programs, designed to control the operation and extend the processing capability of a computer system. In general, a computer's system software performs one or more of the following functions:

- Supports the development of other application software.
- Supports the execution of other application software.
- Monitors the effective use of various hardware resources, such as CPU, memory, peripherals, etc.
- Communicates with and controls the operation of peripheral devices, such as printer, disk, tape, etc.

Hence, system software makes the operation of a computer system more efficient and effective. It helps the hardware components work together, and provides support for the development and execution of application software. The programs included in a system software package are called system programs, and the programmers who prepare system software are referred to as system programmers.

Some of the most commonly known types of system software are:

Operating Systems: Every computer has operating system software, which takes care of the effective and efficient utilization of all the hardware and software components of the computer system.

Programming Language Translators: Programming language translators are system software, which transform the instructions prepared by programmers in a programming language, into a form, which can be interpreted and executed.

Utility Programs: Utility programs are set of programs, which help users in system maintenance tasks, and in performing tasks of routine nature. Some of the task commonly performed by utility programs include formatting of hard disk or floppy disks, taking backup of files stored on hard disk, sorting of the records stored in a file, etc.

2. Application Software:

Application software is a set of one or more programs, designed to solve a specific problem, or to do a specific task. For example, application software for payroll processing products pay slips as the major output, and application software for processing examination results produces mark sheets as the major along with some other statistical reports. Similarly, a program written by a scientist to solve his/her particular research problem is also application software. The programs included in application software package are called application programs, and the programmers who prepare application software are referred to as application programmers. There is literally millions of application software for a wide range of application, ranging from simple applications, such as word processing, inventory management, preparation of tax returns, banking, hospital administration, insurance, publishing, to complex scientific and engineering applications etc. Some of the most commonly known application software's are:

Word-Processing Software: Word processing software enables us to make use of a computer system for creating, editing, and viewing, formatting, sorting, retrieving and printing documents.

Spreadsheet Software: Spreadsheet software is a numeric data analysis tool, which allows us to create a kind of computer ledger. A manual ledger is a book having rows and columns, which accountants use for financial transactions, and for preparing financial statements.

Database Software: A database is a collection of related data stored and treated as a unit for information retrieval purposes. Database software is a set of one or more programs, which enable us to create a database, maintain it, organize its data in desired fashion, and to selectively retrieve useful information from it.

Graphics Software: A graphics software enables us to use a computer system for creating, editing, viewing, sorting, retrieving and printing designs, drawings, pictures, graphs, and anything else that can be drawn in the traditional manner.

Personal Assistance Software: Personal assistance software allows us to use personal computers for sorting

and retrieving our personal information, and planning and managing our schedules, contacts, financial and inventory of important items.

Education Software: Education software allows a computer system to be used as teaching and learning tool. A few examples of such applications are those that teach young children to learn Mathematics; to recognize alphabets; and read whole and sentences, etc.

Programming Languages:

Normally, language is a means of communication, through which any entity can share its feelings, emotions, to each other's i.e. the communication between two parties whether they are machine or human beings, always need a common language. The language used to communicate with the computer is known as programming language. A programming language consists of all the symbols, characters and set of rules that permit people to communicate with computer. To perform a particular task in the computer the programmer writes a sequence of instructions, called program. An instruction is a command given to the computer to perform a certain specified task on a given data.

A language is a means of communication. In our daily life we use various kinds of language such as Nepali, English, Sanskrit, Hindi, etc to communicate our ideas and emotions to others. Similarly a computer language is used by a programmer to instruct a computer what s/he wants it to do. A language that is acceptable to a computer system is called a computer language or programming language, and the process of writing instructions is called programming or coding.

A programming language is an artificial language designed to express computations that can be performed by a machine, particularly a computer. Programming languages can be used to create programs that control the behavior of a machine, to express algorithms precisely, or as a mode of human communication. A programming language is a notation for writing programs, which are specifications of a computation or algorithm.

Comparison between natural language and computer language

Natural Language	Computer language
It is used to communicate between human beings.	It is used to communicate between human beings and computer.
It needs a rich and large vocabulary to enable human beings to express their any thoughts, feelings and ideas.	It has limited vocabulary or restricted vocabulary (FORTRAN, BASIC has around 45)
Each natural language has a systematic method of using the words and symbols of that language, which is defined by the grammar rules of the language.	The word and symbols of a computer language must also used as per set rules, which are known as syntax of the language.
In case of natural languages people can use poor or incorrect vocabulary and grammar and still make them understood.	In case of computer language, we must stick to the exact syntax rules of the language, otherwise it can't be understood, because no computer is capable of correcting and deducing meanings from incorrect instructions.
There are different kinds of natural language, according to the persons, geographical area etc.	Computer languages are classified into 3 groups: Machine level language, assembly language, and High level language.

Classification of Computer Language

There are three type of computer language and they are: machine level language, low level language and high level language.

Machine level language

- A computer understands information's composed of only zeros and hence, it uses binary digits for its operation. The computer's instructions are coded and stored in the memory in the forms of 0's and 1's. a program written in the forms of 0's and 1's is called machine level language.
- The computer language understood by the computer without using translating program is known as machine level language.
- The machine language of computer is normally written as string of binary 0's and 1's.
- A machine language instruction normally has 2 parts format: opcode (operation code) and operand.

Opcode (operation code)	Operand (Address or Location)
----------------------------	----------------------------------

The opcode tells the computer what function to perform and the operand tells the computer where to find or store the data or other instructions, which one to be manipulated. Hence, each instruction tells the computer what operation to perform, and the length and locations of the data fields, which are involved in the operation. Every computer has a set of operation codes, called its instruction set.

Advantage

- Programs written in such language can be executed very fast by the computer. Because instructions are directly understood by the computer and no translation of the program is required.

Disadvantage

- **Machine dependent**
The internal design of every type of computer is different from every other type of computer, so machine language also differs from computer to computer. The program written for one processor (Intel) cannot be run in other processor (Motorola, via etc). The programmer would have to rewrite all the existing programs i.e. it is machine dependent language.
- **Difficult to program**
It is necessary for the programmer either to memorize the dozens of operation code, numbers for the commands in the machines instruction set or to constantly refer to a preference card. A programmer has to keep track of the storage locations of data and instructions. He must be expert about the hardware structure of the computer. So it is difficult to program the machine language.
- **Error prone**
A programmer has to remember the opcodes and must keep track of the storage locations of data and instructions. This task is very difficult for a programmer to concentrate fully on the logic of the program. This frequently results in programming errors.
- **Difficult to modify**
It is difficult to correct or modify machine level programs. Checking machine instructions and to locate errors is very difficult and time consuming process. Similarly, modifying a machine level program later is so difficult
So it is rarely used today

Assembly Language

A language which allows instructions and storage location to be represented by letters and symbols (mnemonics) is called an assembly language or symbolic language. A program written in an assembly language is called assembly language program a symbolic program. It is similar to machine level language, mnemonics codes are used instead of binary digits. Mnemonics used: ADD, SUB, MOV, JMP, HALT, CLA (clear and add), STA (store A register)

Advantage

- **Easier to understand and use**
Assembly language programs are much easier to understand and use, because mnemonics are used instead of numeric opcodes and names for data locations
- **Easier to locate and correct errors**
Programmers need not keep track of storage locations of the data and instructions as well as mnemonics and used, fewer errors are made and those that are easier to find and correct. Additionally assemblers are designed to automatically detect and report errors of use of an invalid mnemonics opcode or a name that has never been defined.
- **Easier to modify**
Since they are easier to understand, it is easier to locate, correct and modify instructions of an assemble language program as compared to machine language.
- **No worry about address**
While writing an assembly language program, programmer need not keep track of storage locations of data and instructions.

Disadvantage

- **Machine dependent**
Assembly language differs from computer to computer, and an assembly language program can be executed only on the computer in which the assembly language has been written. Hence we have to convert all the existing assembly instructions into new computer, which required learning of new language.
- **Knowledge of hardware required**
A programmer must have good knowledge of the characteristic and logical structure of his/her computer to write an efficient and good assembly language programs, because it is machine dependent language.
- **Machine level coding**
Since all instructions is substituted for one machine level instruction. So, writing assembly language program is still time consuming and not very easy.

High level language

The language written in English like structure and each of them has syntax. Such language is written using English words and familiar mathematical symbols and expressions.

Advantage

- **Machine independencies**
A program written in a HLL can be executed on many different kinds of computer.
- **Easier to learn and use**
HLL are easier to learn because they very similar to the natural languages used by us in our daily life. Programmer need not know the internal details of the computer for programming in a HLL i.e. they are easier to use.
- **Fewer error**
Programmer need not worry about how and where to store the instructions and data of the program. This allows the programmer to concentrate more on the logic of the program under development. All these factors lead to fewer programming errors during program development. Furthermore, compiler and interpreters are designed automatically detect and indicate syntax error.
- **Lower program preparation cost**
Writing programs in HLL requires less time and effort, which ultimately leads to lower program preparation cost.
- **Better documentation**
The statements of a program written in HLL are very similar to the natural language statements. Hence they can be easily understood by a programmer familiar with the problem domain. As a result

very few or no separate comments statements are required I program written in HLL. Due to this reason, HLL are also sometimes referred to as self-documenting language.

- **Easier to maintain**

These languages are easier to understand, and hence it is easier to locate, correct and modify instructions as and when desired. Without any compilation, programmer can insert or remove any certain instructions from program i.e. major changes can be incorporated with very little effort.

Disadvantage:

- **Lower efficiency**

Programs written in HLL take more time to execute and require more main memory space. Hence, program written in HLL are less efficient than others.

- **Less flexibility**

HLL don not have instruction or mechanism to control the computers CPU, memory and register. An assembly language provides the programmer access to all the especial features of the machine they are using. Certain types of operations, which are easily programmed using the machines assembly language, are impractical to attempt using a HL. Hence, HLL are less flexible than assembly language.

Traditional and Structural Programming Concept

Structured programming is subsets of procedural programming that implements a logical structure on the program being written to make it more efficient and easier to understand and modify. In procedural programming languages, the program instructions are written in a sequence or in a specific order in which they must be executed to solve a specific problem. It means that the order of program instructions is very important. It is also known as modular programming concept.

Structured programming frequently employs a top-down design model, in which developers map out the overall program structure into separate subsections. A defined function or set of similar functions is coded in a separate module or sub-module, which means that code, can be loaded into memory more efficiently and that modules can be reused in other programs. After a module has been tested individually, it is then integrated with other modules into the overall program structure.

Example

ALGOL, Pascal, PL / I, Ada, C, etc

Comparison between structured programming and object oriented programming

Structured Programming	Object-Oriented Programming
Top-down approach is followed.	Bottom-up approach is followed.
Focus is on algorithm and control flow.	Focus is on object model.
Program is divided into a number of sub-modules, or functions, or procedures.	Program is organized by having a number of classes and objects.
Functions are independent of each other.	Each class is related in a hierarchical manner.
Function call is used.	Message passing is used.
It is less secure as there is no way of data hiding.	It is more secure as having data hiding feature.
Maintenance is costly.	Maintenance is relatively cheaper.
No encapsulation. Data and functions are separate.	Encapsulation packages code and data altogether. Data and functionalities are put together in a single entity.
Example: C, ADA, Pascal, etc	Example: C++, C#, Java, Visual Basic.net, Ruby, etc